

A candidate proof of Problem 6.4 in Colour-balanced subgraphs

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

The requested $O(k^2)$ bound improves to $O(k \log(k+1))$ for Hamilton cycles in both complete and complete bipartite graphs, including for bounded vector labels in an arbitrary norm. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Problem 6.4](#) from *Colour-balanced subgraphs*.

1.1 Problem statement

Problem. Improve the bounds on $f_h(H)$ for a Hamilton cycle H in complete or complete bipartite graphs.

1.2 Catalog context

The general upper bound from Theorem 1.2 gives $O(k^2)$ for Hamilton cycles in complete graphs, and the same bound holds for complete bipartite graphs by the paper’s methods. A necklace-splitting argument shows any improvement to the bipartite matching bound yields an equal improvement here, so progress on either problem benefits the other.

1.3 Claimed result

Model verdict: **proved** (confidence: high).

The requested $O(k^2)$ bound improves to $O(k \log(k+1))$ for Hamilton cycles in both complete and complete bipartite graphs, including for bounded vector labels in an arbitrary norm.

The model’s own caveat: Optimality and priority relative to later revisions of the source paper are not established.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 An $O(k \log k)$ bound

The earlier attempt’s midpoint-splicing and approximate-convexity arguments are sound. Below I verify both, sharpen the midpoint estimate, and extend the complete-graph argument to arbitrary

bounded vector labels. Thus the improvement does not depend on interpreting f_h only as a recolouring statistic.

Here “proved” refers to the requested improvement in the upper bound, not to determining its optimal order.

2.1.1 1. Precise statement

For $d \geq 1$, put

$$B_d = \lceil \log_2 d \rceil + 2.$$

If h labels the edges of a graph by vectors, write

$$h(F) = \sum_{e \in E(F)} h(e).$$

2.1.1.1 Theorem 1 Let X be a real d -dimensional normed space, and suppose that

$$h : E(G) \longrightarrow X, \quad \|h(e)\| \leq L.$$

1. If $G = K_{n,n}$, where $n \geq 2$, then G has a Hamilton cycle H satisfying

$$\left\| h(H) - \frac{2}{n} h(G) \right\| \leq 6LdB_d. \quad (1)$$

2. If $G = K_N$, where $N \geq 3$, then G has a Hamilton cycle H satisfying

$$\left\| h(H) - \frac{2}{N-1} h(G) \right\| \leq 6LdB_d + 8Ld + 2L. \quad (2)$$

The centring vectors in (1) and (2) are the expected label sums of uniformly random Hamilton cycles. In particular, for unit-bounded, zero-sum d -dimensional labels, the least norm of a Hamilton-cycle label sum is $O(d \log(d+1))$.

For the recolouring formulation, the theorem gives the following explicit bounds.

2.1.1.2 Corollary 2 Let $k \geq 2$, and suppose that the host graph is k -edge-coloured with equally large colour classes.

- If the host is $K_{n,n}$, $n \geq 2$, and $k \mid 2n$, some Hamilton cycle can be made colour-balanced by recolouring at most

$$3kB_k \quad (3)$$

of its edges.

- If the host is K_N , $N \geq 3$, and $k \mid N$, some Hamilton cycle can be made colour-balanced by recolouring at most

$$3kB_k + 4k + 1 \quad (4)$$

of its edges.

Consequently, the quadratic upper bound in the question is replaced by

$$\boxed{O(k \log(k+1))}.$$

The substantive ingredient is an improved vector-labelled bipartite matching bound.

2.1.2 2. Halving a vector-valued necklace

We first establish the continuous splitting fact needed in the matching argument.

2.1.2.1 Lemma 3 Let $f : [0, r] \rightarrow \mathbb{R}^d$ be integrable. There is a set $A \subseteq [0, r]$ whose indicator is constant between at most d cut points and such that

$$\int_A f(x) dx = \frac{1}{2} \int_0^r f(x) dx. \quad (5)$$

2.1.2.1.1 Proof For $x = (x_0, \dots, x_d) \in S^d$, partition $[0, r]$ into consecutive intervals $J_0, \dots, J_d$ of lengths

$$rx_0^2, \dots, rx_d^2.$$

Define

$$F(x) = \sum_{i=0}^d \operatorname{sgn}(x_i) \int_{J_i} f(t) dt.$$

The map $F : S^d \rightarrow \mathbb{R}^d$ is continuous: when a coordinate tends to zero, its interval has vanishing length, so the corresponding integral tends to zero. It is also odd.

By the Borsuk–Ulam theorem, F has a zero. Taking A to be the union of the positively signed intervals gives (5). There are at most d internal cut points. $\square$

No positivity assumption on the coordinates of f is needed.

2.1.3 3. Approximate midpoint closure for matching sums

2.1.3.1 Lemma 4 Suppose that the edges of $K_{r,r}$ are labelled by vectors in X , each of norm at most L . For any two perfect matchings P, Q , there is a perfect matching R such that

$$\left\| h(R) - \frac{h(P) + h(Q)}{2} \right\| \leq 3Ld. \quad (6)$$

2.1.3.1.1 Proof Let the bipartition be $U \cup V$. Regard P, Q as bijections from V to U , and define a permutation σ of V by

$$Q(v) = P(\sigma(v)).$$

Order the vertices of V by concatenating cyclic orders of the cycles of σ . Write the resulting order as $v_1, \dots, v_r$. Let P_i, Q_i denote the respective matching edges incident with v_i .

On $I_i = [i-1, i]$, put

$$f(x) = h(P_i) - h(Q_i).$$

Apply Lemma 3, after identifying X linearly with $\mathbb{R}^d$, and set

$$\alpha_i = |A \cap I_i|.$$

Then

$$\sum_i \alpha_i (h(P_i) - h(Q_i)) = \frac{1}{2} \sum_i (h(P_i) - h(Q_i)). \quad (7)$$

Choose $s_i \in \{0, 1\}$ nearest to α_i , breaking ties arbitrarily. At most d of the numbers α_i lie strictly between zero and one, since each such unit interval contains an internal cut point. Therefore

$$\sum_i |s_i - \alpha_i| \leq \frac{d}{2}. \quad (8)$$

Select P_i if $s_i = 1$, and Q_i if $s_i = 0$. This gives one selected edge at every vertex of V , though it need not yet be a matching. Let z be its label sum. By (7) and (8),

$$\begin{aligned} \left\| z - \frac{h(P) + h(Q)}{2} \right\| &= \left\| \sum_i (s_i - \alpha_i)(h(P_i) - h(Q_i)) \right\| \\ &\leq 2L \sum_i |s_i - \alpha_i| \leq Ld. \end{aligned} \quad (9)$$

We now bound the number of matching defects.

For each i , the chosen value s_i occurs as the indicator of A at some non-cut point in the interior of I_i . Choosing such points in increasing order shows that the linear binary sequence

$$s_1, \dots, s_r$$

has at most d changes.

Consider one cycle block of σ , and let c be its number of internal linear changes. Closing that block cyclically adds at most one change; if it adds one, then $c \geq 1$. Hence the number of cyclic changes in that block is at most $2c$. Summing over all blocks, the total number b of cyclic changes satisfies

$$b \leq 2d. \quad (10)$$

For a cycle written so that $\sigma(v_i) = v_{i+1}$, the degree of $P(v_i)$ in the selected edge set is

$$1 + s_i - s_{i-1},$$

with cyclic subscripts. Consequently, there are exactly $b/2$ vertices of U of degree two and $b/2$ of degree zero.

Delete one selected edge at each degree-two vertex. This frees $b/2$ vertices of V . Match them arbitrarily to the degree-zero vertices of U , using completeness of the host. The resulting graph is a perfect matching R , obtained by replacing $b/2 \leq d$ selected edges. Thus

$$\|h(R) - z\| \leq 2L \frac{b}{2} \leq 2Ld. \quad (11)$$

Combining (9) and (11) proves (6). $\square$

The important feature is that the error is independent of r .

2.1.4 4. Approximate convexification costs only a logarithm

2.1.4.1 Lemma 5 Let S be a finite subset of a d -dimensional normed space. Suppose that for every $u, v \in S$, some $w \in S$ satisfies

$$\left\| w - \frac{u + v}{2} \right\| \leq D. \quad (12)$$

Then every $x \in \text{conv}(S)$ satisfies

$$\text{dist}(x, S) \leq DB_d. \quad (13)$$

2.1.4.1.1 Proof By Carathéodory's theorem, write

$$x = \sum_{i=1}^m \beta_i x_i, \quad x_i \in S, \quad m \leq d + 1.$$

The case $m = 1$ is immediate.

First suppose that all coefficients have a common dyadic denominator 2^q . Form a complete binary tree with 2^q leaves, placing the required copies of $x_1, \dots, x_m$ in consecutive blocks.

Assign representatives in S to the internal nodes, proceeding upwards. If all leaves below a node are equal, choose that common point. Otherwise use (12) on its two child representatives.

At depth j from the root, the number H_j of nodes whose leaves are not all equal is at most

$$H_j \leq \min(2^j, m - 1). \quad (14)$$

Indeed, their disjoint leaf intervals must each contain a boundary between consecutive blocks.

An error introduced at depth j contributes to the root with coefficient 2^{-j} . Therefore the root representative $z \in S$ satisfies

$$\begin{aligned} \|z - x\| &\leq D \sum_{j=0}^{q-1} 2^{-j} H_j \\ &\leq D \sum_{j \geq 0} \min(1, (m - 1)2^{-j}) \quad (15) \\ &\leq D(\lceil \log_2(m - 1) \rceil + 2) \\ &\leq DB_d. \end{aligned}$$

For arbitrary coefficients, approximate them by dyadic coefficients summing to one. Since S is finite, a subsequence of the resulting root representatives is constant. Passing to the limit proves (13). $\square$

This is where the quadratic dependence is avoided: a sequential convexification can pay an $O(d)$ error $O(d)$ times, whereas the weighted dyadic construction pays only $O(\log d)$ levels.

2.1.4.2 Proposition 6: vector matching bound For an edge-labelling of $K_{r,r}$ in X with $\|h(e)\| \leq L$, there is a perfect matching M such that

$$\left\| h(M) - \frac{1}{r} h(K_{r,r}) \right\| \leq 3LdB_d. \quad (16)$$

In fact, every point in the convex hull of the perfect-matching label sums is within $3LdB_d$ of a perfect-matching label sum.

2.1.4.2.1 Proof Apply Lemma 5 to

$$S = \{h(M) : M \text{ is a perfect matching}\},$$

using Lemma 4 with $D = 3Ld$.

A uniformly random perfect matching contains each edge with probability $1/r$, so

$$\frac{1}{r} h(K_{r,r}) \in \text{conv}(S).$$

This proves (16). $\square$

2.1.5 5. Hamilton cycles in complete bipartite graphs

Let the parts of $K_{n,n}$ be

$$A = \{a_1, \dots, a_n\}, \quad B,$$

with A cyclically ordered and $a_{n+1} = a_1$.

Construct an auxiliary $K_{n,n}$ between the gaps

$$g_i = (a_i, a_{i+1})$$

and the vertices of B . Give its edges the labels

$$\lambda(g_i b) = h(a_i b) + h(a_{i+1} b). \quad (17)$$

These labels have norm at most $2L$.

A perfect matching in the auxiliary graph assigns distinct vertices b_i to the gaps and produces the Hamilton cycle

$$a_1 b_1 a_2 b_2 \cdots a_n b_n a_1. \quad (18)$$

Its h -sum is exactly the auxiliary matching's λ -sum.

Each original edge appears in exactly two auxiliary labels. Hence

$$\frac{1}{n} \sum_{i,b} \lambda(g_i b) = \frac{2}{n} h(K_{n,n}). \quad (19)$$

Applying Proposition 6 with label bound $2L$ gives (1).

This construction also works for $n = 2$; it produces the unique four-cycle.

2.1.6 6. Hamilton cycles in complete graphs

We need a nearly balanced cut, now for arbitrary normed vector labels.

2.1.6.1 Lemma 7 Let $N \geq 4$, and suppose the edges of K_N have labels in X of norm at most L . Put

$$a = \left\lceil \frac{N}{2} \right\rceil, \quad b = \left\lfloor \frac{N}{2} \right\rfloor, \quad p = \frac{2ab}{N(N-1)}.$$

There is a partition $V(K_N) = A \cup B$, with $|A| = a$, $|B| = b$, such that

$$\|h(E(A, B)) - p h(K_N)\| \leq 2LdN. \quad (20)$$

2.1.6.1.1 Proof Choose A uniformly among the a -subsets, and let I_e indicate that e crosses the partition. Thus $\mathbb{E} I_e = p$.

We first record a scalar variance bound. Let $|w_e| \leq L$, and put

$$Z = \sum_e w_e (I_e - p).$$

For adjacent distinct edges e, f ,

$$\mathbb{P}(I_e = I_f = 1) = \frac{p}{2},$$

and therefore

$$|\text{Cov}(I_e, I_f)| = \left| p \left(\frac{1}{2} - p \right) \right| \leq \frac{1}{N}. \quad (21)$$

For disjoint edges, direct enumeration gives

$$\text{Cov}(I_e, I_f) = \begin{cases} \frac{N}{2(N-1)^2(N-3)}, & N \text{ even,} \\ \frac{N+1}{2N^2(N-2)}, & N \text{ odd.} \end{cases} \quad (22)$$

Both are at most $4/N^2$. For example, the joint crossing probability before subtracting p^2 is

$$\frac{4a(a-1)b(b-1)}{N(N-1)(N-2)(N-3)}.$$

There are $\binom{N}{2}$ edges, $3\binom{N}{3}$ unordered adjacent edge pairs, and $3\binom{N}{4}$ unordered disjoint edge pairs. Using absolute values for the covariance terms,

$$\begin{aligned} \text{Var } Z &\leq L^2 \left(\frac{1}{4} \binom{N}{2} + \frac{2}{N} 3 \binom{N}{3} + \frac{8}{N^2} 3 \binom{N}{4} \right) \\ &\leq \frac{17}{8} L^2 N^2 < 4L^2 N^2. \end{aligned} \quad (23)$$

Choose a basis $u_1, \dots, u_d$ of X with

$$\|u_j\| = 1$$

and coordinate functionals φ_j of dual norm one. Such a basis exists: maximize the absolute determinant of d vectors in the unit ball. Replacing a basis vector by any other unit-ball vector shows that every corresponding coordinate has absolute value at most one. Also,

$$\|x\| \leq \sum_{j=1}^d |\varphi_j(x)|. \quad (24)$$

Set

$$T = \sum_e (I_e - p) h(e).$$

Applying (23) to the scalar weights $\varphi_j(h(e))$, which have absolute value at most L , gives

$$\begin{aligned} \mathbb{E}\|T\| &\leq \sum_{j=1}^d \mathbb{E}|\varphi_j(T)| \\ &\leq \sum_{j=1}^d \sqrt{\mathbb{E}[\varphi_j(T)^2]} \\ &\leq 2LdN. \end{aligned}$$

Some partition therefore satisfies (20). $\square$

2.1.6.2 Completing the proof of Theorem 1 For $N = 3$, the Hamilton cycle consists of all three edges, and the expression on the left of (2) is zero. Assume henceforth that $N \geq 4$.

Choose A, B as in Lemma 7, cyclically order

$$A = \{a_1, \dots, a_a\},$$

and let

$$T = h(E(A, B)), \quad S = h(K_N).$$

If N is even, $|B| = a$. If N is odd, adjoin a dummy symbol $*$ to B , obtaining a set B^* of size a .

Construct an auxiliary $K_{a,a}$ between the cyclic gaps $g_i = (a_i, a_{i+1})$ and B^* . For a real vertex $v \in B$, set

$$\lambda(g_i v) = h(a_i v) + h(a_{i+1} v). \quad (25)$$

In the odd case, set

$$\lambda(g_i *) = h(a_i a_{i+1}). \quad (26)$$

All auxiliary labels have norm at most $2L$.

Every auxiliary perfect matching produces a Hamilton cycle: insert a real vertex into its assigned gap, and in the odd case use the direct edge across the dummy-assigned gap.

Let μ be the mean auxiliary perfect-matching label sum. Then

$$\mu = \begin{cases} \frac{2}{a}T, & N \text{ even,} \\ \frac{2}{a}T + \frac{1}{a}Y, & N \text{ odd,} \end{cases} \quad (27)$$

where, in the odd case,

$$Y = \sum_{i=1}^a h(a_i a_{i+1}).$$

If N is even, then

$$\frac{2p}{a} = \frac{2}{N-1},$$

so

$$\mu - \frac{2}{N-1}S = \frac{2}{a}(T - pS). \quad (28)$$

If N is odd, then $2p/a = 2/N$, and consequently

$$\mu - \frac{2}{N-1}S = \frac{2}{a}(T - pS) + \frac{Y}{a} - \frac{S}{\binom{N}{2}}. \quad (29)$$

Both Y/a and $S/\binom{N}{2}$ have norm at most L . Using Lemma 7 and $a \geq N/2$, we obtain in either case

$$\left\| \mu - \frac{2}{N-1}S \right\| \leq 8Ld + 2L. \quad (30)$$

By Proposition 6, an auxiliary perfect matching has label sum within

$$6LdB_d$$

of μ . Its corresponding Hamilton cycle therefore satisfies (2). This completes the proof of Theorem 1. $\square$

2.1.7 7. Translation to colour balance

Give every edge of colour j the label

$$h(e) = e_j \in (\mathbb{R}^k, \|\cdot\|_1).$$

Thus every edge label has norm one, and $h(F)$ is the colour-count vector of F .

For a globally colour-balanced $K_{n,n}$,

$$\frac{2}{n}h(K_{n,n}) = \frac{2n}{k}\mathbf{1}.$$

For a globally colour-balanced K_N ,

$$\frac{2}{N-1}h(K_N) = \frac{N}{k}\mathbf{1}.$$

If a subgraph F has kq edges and colour counts $m_1, \dots, m_k$, its minimum recolouring number is

$$\rho(F) = \sum_{j=1}^k (m_j - q)_+ = \frac{1}{2} \sum_{j=1}^k |m_j - q|. \quad (31)$$

Indeed, every surplus edge must be recoloured, and recolouring surplus edges into deficient colours attains this number.

Taking $d = k$, $L = 1$, and halving the bounds in Theorem 1 proves (3) and (4).

2.1.8 Conclusion and scope

The proof establishes, uniformly in the host order,

$$f_h(H) = O(k \log(k+1))$$

in the bounded, zero-sum vector-label formulation, as well as the explicit recolouring bounds above. It also establishes the corresponding $O(k \log(k+1))$ bipartite perfect-matching estimate.

Thus it answers the stated request for an improvement over $O(k^2)$. It does **not** establish an $O(k)$ bound or determine the optimal order. I have not independently checked whether a later revision of the source already contains this improvement.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2604.09449__03
Catalog entry	Problem 6.4
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Colour-balanced subgraphs
Model verdict	proved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2604.09449__03/output.md
Independent review	None
arXiv	arXiv:2604.09449
Previous attempt	attacks/2604.09449__03: gpt-5.6-sol verdict partial

Model's caveats	Optimality and priority relative to later revisions of the source paper are not established.
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The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/2604.09449__03/`.